

# Fault Detection In Transmission Lines using Summation of Squared Three-Phase Currents

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**Abstract**—Detection, classification, and identifying location by distance protection are the three stages of fault identification in transmission lines. When fault detection is quick, the entire protection procedure can be completed quickly. In this paper, a fault detection approach based on the summation of squared three-phase currents (SSC) and a moving average scheme is presented. The SSC signal has a constant value in normal conditions and shows significant variation when it is faulty. The fluctuation in the SSC signal is discovered, and the fault is detected using fault detection criteria (FDC). This fault detection approach helps to prevent accidental tripping and ensures the reliability of the power system. This method examines only one signal which is the sum of three phase current signals. This decreases the computational effort and makes this a cost-effective technique. Furthermore, fault detection takes only a few milliseconds using this technology. MATLAB SIMULINK software is used to test this technique in an IEEE 6-bus system.

**Index Terms**—Transmission line (TL), summation of squared three phase currents (SSC), fault detection criteria (FDC).

## I. INTRODUCTION

Transmission lines are an unavoidable aspect of the power system, as they transfer energy from generators to consumers. Due to diverse environmental circumstances, there will be several disruptions and contingencies in its journey. That is, transmission line protection is adequate for reliable functioning of the power system. There are various transmission line protection systems available, the most popular and effective of which is distance protection. Distance protection is usually divided into three stages: detection, classification, and fault location identification[1]. The protection mechanism becomes faster as the speed of detection improves. In addition, the fault detection stage gathers the necessary data for the next phases. As a result, our goal is to improve the speed with which faults are detected.

Signal processing, travelling wave, phasor, and artificial intelligence-based technologies are only a few of the fault detection techniques available. There are also some more strategies such as correlation coefficient, sequential component, and fuzzy logic based methods. The numerous approaches for detecting a fault in a transmission line that are now available are shown in Fig 1. Each strategy has its own set of benefits and drawbacks.

Signal processing-based approaches in Group A include wavelet transform-based methods [2] and Fourier transform-

based methods [3]. Voltage rms-based methods, current rms-based methods, and voltage and current rms-based methods are among the phasor-based techniques in Group B [4]. The AI approaches in Group C include artificial neural networks [5], support vector machines [6], and decision trees. Group D [7], [8] includes techniques based on travelling waves. Other techniques not covered by the other groups are combined together in Group E, which is termed as miscellaneous. This area includes fuzzy logic-based methods, correlation-based methods, sequential component-based methods, and so on [9].

The fundamental disadvantage of group A is its high computational cost, and the parameters of this technique should be re-adjusted to perform well for different systems. Furthermore, they require a high sample frequency to achieve good results. All of the characteristics of group B are determined by the phasor estimation method chosen. Their performance suffers greatly if the method is not chosen properly. This group is not a better approach for detection based on the comparison offered. Although AI-based systems are capable of detecting faults efficiently, one of the key downsides of these methods is the significant amount of training effort required for optimal performance. The group C requires trained operators for secure operation. The techniques employed in group D are fast and accurate. But they require measurement devices of high sampling rate. Even though computational burden and sampling rate are low for group E techniques, the speed of fault detection is relatively low when compared to other techniques. For efficient performance, group C requires trained operators. The procedures used in group D are quick and precise. However, they necessitate high-sampling-rate measurement instruments. Group E approaches have a low computational effort and sample rate, but their fault detection time is high when compared to other techniques.

All of the fault detection methods listed above have their own set of benefits and drawbacks. As a result, an accurate, dependable, and quick problem detection mechanism is sufficient for improving power transmission in an electric power system. A new idea based on moving average and three phase currents is introduced [10] to detect the faults in TL. The summation of squared three phase currents (SSC) signal is calculated as the first step in this method. The SSC signal is then subjected to a 10 ms moving window with a 1 ms sliding time step

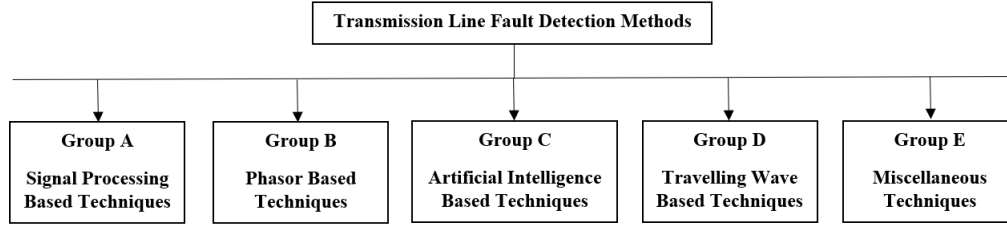


Fig. 1. Fault detection methodologies

as the moving average. During normal system operation, the SSC signal is constant, but it varies greatly when the system is in a fault condition. Different problems can be recognised and thus the power system's reliability can be maintained by examining the amount of variations.

This approach, which uses a moving average and an SSC signal, can accurately detect a problem within a few milliseconds of its onset. To reduce computational burden, only one signal is processed, which is a mixture of three phase currents. This technique is also cost effective because there is no need for a potential transformer. This method can be directly applied to any system. There is no need of readjustment for different network structures. The fault detection speed is fast enough to identify and clear the fault as quickly as possible.

The new fault detection approach is described in detail in Section II of the paper. The simulation results obtained in different conditions are presented in Section III. The new method is compared to existing fault detection methods in section IV. Finally, the conclusions are presented in Section V.

## II. SSC BASED FAULT DETECTION METHOD

The block diagram of the fault detection system is shown in Fig. 2. Firstly, sample all the three phase currents and square their values. All the squared values are combined together to form a single signal referred to as the SSC signal. In the further processing steps, only this signal is used to detect the fault. Then determine the moving average of the SSC signal. Then a fault detection criterion (FDC) is used to detect whether there is a fault or not.

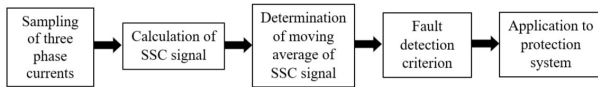


Fig. 2. Block diagram of fault detection system

### A. Method of operation

The summation of squared three phase currents (SSC) of a transmission line is defined as shown below:

$$SSC = i_A^2(t) + i_B^2(t) + i_C^2(t) \quad (1)$$

where  $i_A(t)$ ,  $i_B(t)$ ,  $i_C(t)$  are instantaneous three-phase currents at relay location. This signal is constant during normal

condition. In order to verify that the measured three-phase currents of the TL are as follows:

$$\begin{aligned} i_A(t) &= I_{mA} \cos(\omega t + \phi_A) \\ i_B(t) &= I_{mB} \cos(\omega t + \phi_B - 120^\circ) \\ i_C(t) &= I_{mC} \cos(\omega t + \phi_C + 120^\circ) \end{aligned} \quad (2)$$

where  $I_{mA}$ ,  $I_{mB}$ ,  $I_{mC}$  are the maximum values of three phase currents and  $\phi_A$ ,  $\phi_B$ ,  $\phi_C$  are their phase angles.

1) **Normal Condition:** In normal condition, transmission networks can be assumed to be balanced and symmetrical. The maximum values of three-phase currents and phase angles are nearly equal under this condition, hence the following values are assumed:

$$\begin{aligned} I_{mA} &= I_{mB} = I_{mC} = I_m \\ \phi_A &= \phi_B = \phi_C = \phi \\ \omega t + \phi &= \theta \end{aligned} \quad (3)$$

Thus SSC can be calculated as shown below:

$$SSC_{normal} = I_m^2 \left[ \frac{3}{2} + \frac{\cos(2\theta)}{2} + \frac{\cos(2\theta - 240^\circ)}{2} + \frac{\cos(2\theta + 240^\circ)}{2} \right] \quad (4)$$

The resultant of three sinusoidal terms is zero and therefore

$$SSC_{normal} = \frac{3}{2} I_m^2 \quad (5)$$

Thus, it is verified that the magnitude of SSC signal is constant during normal condition of the system.

2) **Faulty condition:** In most TLs, fault current signals have a dc offset component that may be described using decaying exponential functions. As a result, each phase's fault current signals can be expressed as follows:

$$i(t) = I_0 e^{-\frac{t}{T}} + I'_m \cos(\omega t + \phi') \quad (6)$$

where  $T$  and  $I_0$  are time constant and magnitude of decaying dc offset. Also  $I'_m$  and  $\phi'$  are the maximum value of current and phase angle under fault conditions.

A three-phase symmetrical fault is used as an example of a typical fault in this explanation of SSC signal variation. The decaying dc component is considered to present in all stages. Furthermore, for all phases, the maximum value, phase angle,

dc offset magnitude, and time constant are all regarded to be the same.

$$\begin{aligned} \text{SSC}_{\text{fault}} = & I_m'^2 \cos^2(\theta') + I_m'^2 \cos^2(\theta' - 120^\circ) \\ & + I_m'^2 \cos^2(\theta' + 120^\circ) + 3I_0'^2 e^{-\frac{2t}{\tau}} + 2I_0' I_m' e^{-\frac{t}{\tau}} \\ & [\cos(\theta') + \cos(\theta' - 120^\circ) + \cos(\theta' + 120^\circ)]. \end{aligned} \quad (7)$$

Hence, SSC can be simplified as

$$\text{SSC}_{\text{fault}} = \frac{3}{2} I_m'^2 + 3I_0'^2 e^{-\frac{2t}{\tau}}. \quad (8)$$

A moving average is applied to this signal in order to detect changes in SSC. In a window of length  $T_0$ , the moving average for a discrete signal like  $x_n(t)$  is defined by

$$\text{MA}_n = \frac{1}{T_0} \sum_{n=t-T_0}^t x_n(t) \quad (9)$$

where  $\text{MA}_n$  denotes the moving average of signal  $x(t)$  in the  $n^{\text{th}}$  window. With a window length of 10 ms and a time step of 1ms, the SSC signal is subjected to a moving average window function. The ratio of each window's average for the SSC signal to the previous window's average is defined as fault detection criterion (FDC).

$$\text{FDC}_n = \frac{\text{MA}_n}{\text{MA}_{n-1}}. \quad (10)$$

Under normal conditions, FDC is close to unity. However, it deviates from one in the event of a fault. Because the SSC signal's value remains constant during normal operation, the mean values of two typical successive windows are the same. As a result, their ratio, called FDC, is very near to one. Under this circumstance, small FDC variations from unity indicate the presence of various disturbances in the signals. The fault can be detected by comparing this value to a predetermined threshold. FDC is noticed as a fault if it exceeds the predetermined threshold.

$$\text{FDC}_{nf} > \text{TH} \rightarrow \text{Fault}. \quad (11)$$

The TH shows the threshold value and  $nf$  denotes the window in which fault is detected.

The value of threshold is constant for different systems and here it is taken as 1. The step by step processes done in this approach is portrayed in the Fig. 3.

### III. SIMULATION

In this section, the simulation of IEEE 6-bus system and application of fault detection methodology in it is done. Different types of faults are applied to this system to analyse the variations in the value of FDC and also in SSC signal. The single line diagram of IEEE 6 bus system is shown in Fig. 4. There are three generators and three loads. Fault is applied on line 3-6 and change in both voltage and current are measured. To obtain the waveform of voltage and current, fault is applied between 0.03s and 0.05s. The variations in different fault conditions like LG, LL, LLG and LLLG are obtained.

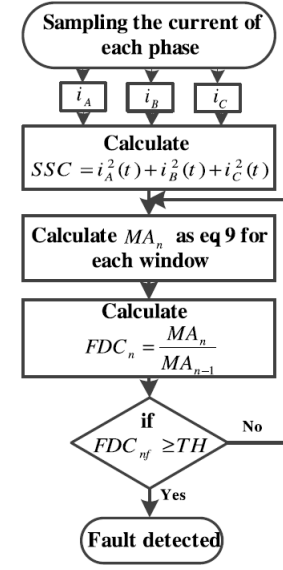


Fig. 3. Flowchart of fault detection methodology

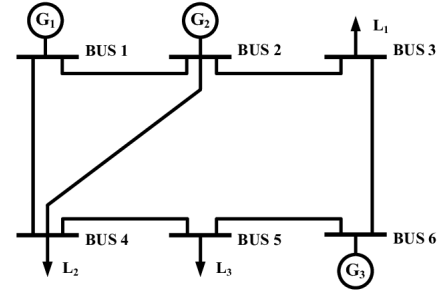


Fig. 4. Single line diagram of IEEE 6 bus system

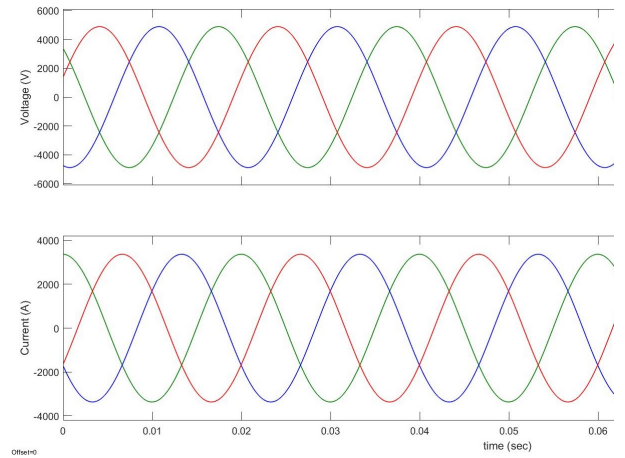


Fig. 5. Three phase voltages and currents in normal condition

The waveform of three phase voltage and current in normal condition is shown in Fig. 5. The voltage and current waveforms in the case of LG, LL, LLG and LLLG fault conditions are portrayed in Fig. 6, Fig. 7 and Fig. 8 and Fig. 9 respectively. The variations in voltage and current waveform

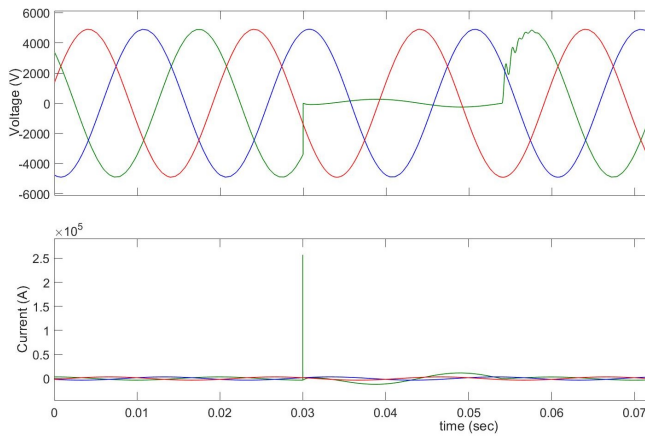


Fig. 6. Three phase voltages and currents in LG fault condition

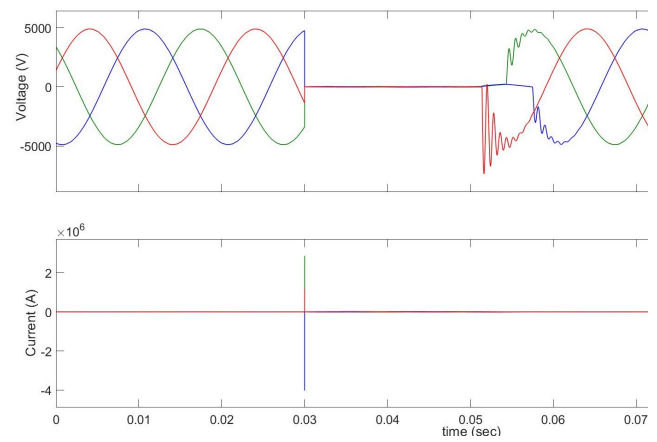


Fig. 9. Three phase voltages and currents in LLLG fault condition

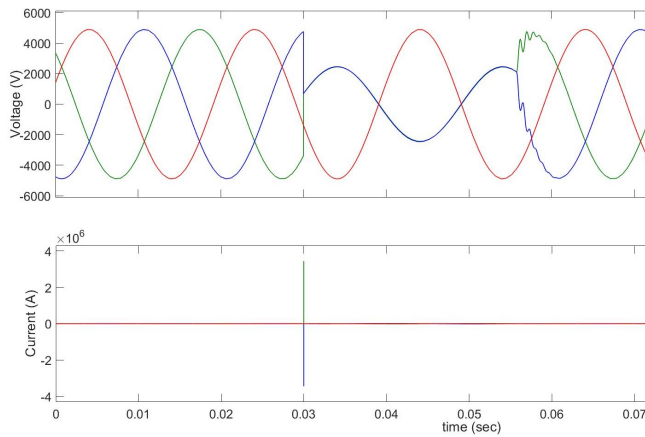


Fig. 7. Three phase voltages and currents in LL fault condition

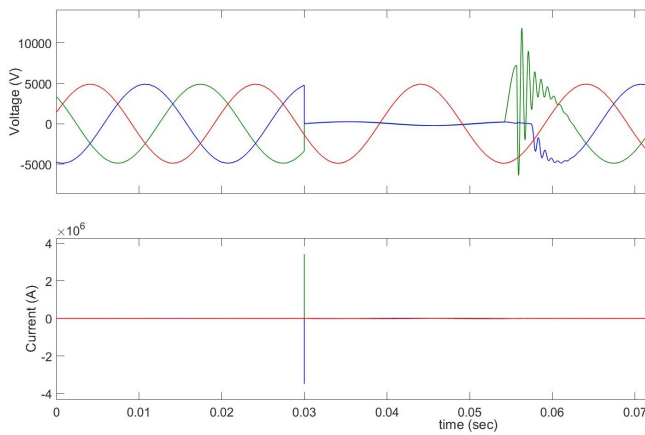


Fig. 8. Three phase voltages and currents in LLG fault condition

is more in the case of LLLG fault condition. The fault current is of the order of  $4 \times 10^6$  amperes in LLLG fault condition.

The SSC signal is constant during normal condition with a value of  $3.5 \times 10^7$  ( $3500^2 \times 3$ ) as shown in Fig. 10. But it experiences changes in different fault conditions. The variations in SSC signal in different fault conditions are observed in Fig.

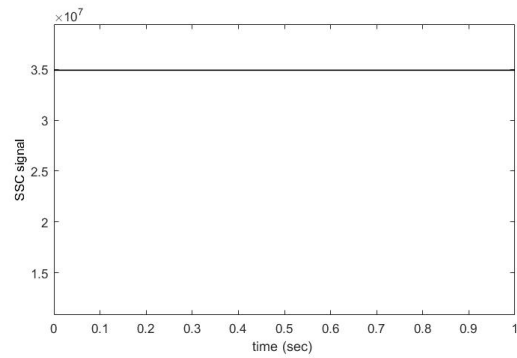


Fig. 10. SSC signal in normal condition

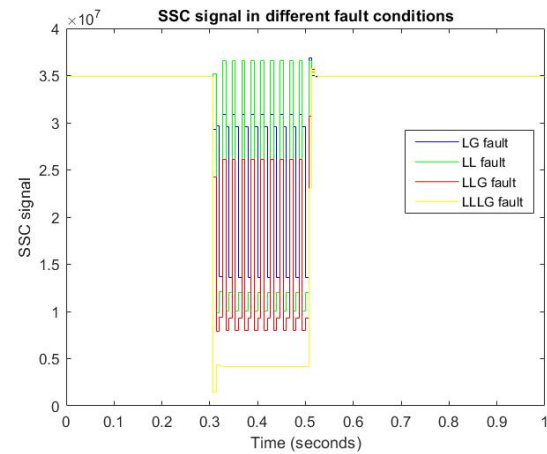


Fig. 11. SSC signal in different fault conditions

11 and the change is more in case of LLLG fault followed by LLG, LL and LG fault conditions.

The value of FDC in normal condition is 1. The variation of FDC value in different types of faults is shown in Fig. 12, Fig. 13, Fig. 14 and Fig. 15 respectively. If FDC is greater than 1, it implies the presence of fault. Here, fault is applied between 0.3s and 0.5s. Upto 0.3s, value of FDC is 1 and from

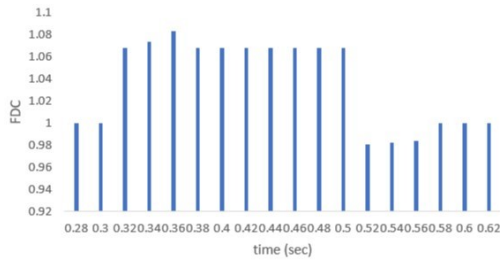


Fig. 12. FDC in LG condition

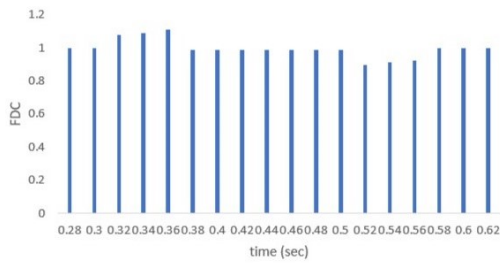


Fig. 13. FDC in LL fault condition

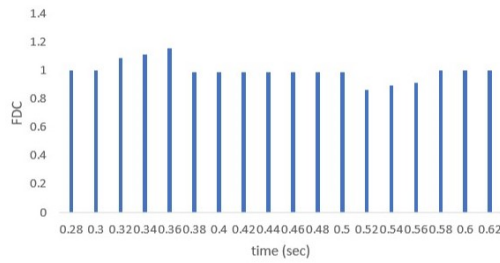


Fig. 14. FDC in LLG fault condition

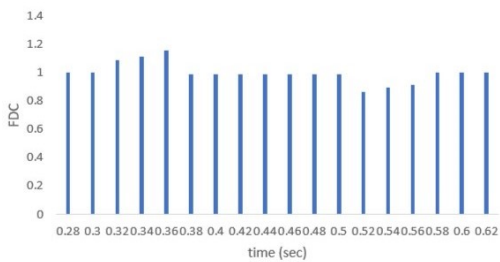


Fig. 15. FDC in LLLG fault condition

0.3s to 0.4s its value is greater than 1. From 0.4s to 0.5s FDC value is close to 1 and it is less than 1 in the interval of 0.5-0.6s. Again from 0.6s onwards the value of FDC remains at 1. The LLLG fault condition has highest FDC value followed by LLG, LL and LG fault conditions.

#### IV. COMPARATIVE ASSESSMENT

In this section, maximum fault detection time of the new method is compared with some other similar approaches. The

methods used for comparison are as follows:

- 1) Sample-to-sample comparison method  
To detect the fault in this method, the difference between the current and prior samples is calculated and compared to a threshold value. In the steady state of the power system, the difference is meant to be a constant, but it rises rapidly in faulty conditions [11].
- 2) Moving sum method  
This approach uses current samples to calculate one cycle summation for a side of TLs. Because of the symmetrical shape of the current waveform, it is observed in power systems. The summation is near to 0 in normal conditions. However, it reaches a preset threshold under the fault condition [12].
- 3) Discrete wavelet transform based method  
The wavelet transform is a valuable technique for analysing transient voltages and currents in power system protections. This approach analyses TL's three-phase currents using a db 4 mother wavelet to provide detail coefficients across a half-cycle moving window. In the moving window, the total of such detail coefficients is compared to a limit. If this value exceeds the limit, the fault is detected [13].
- 4) Difference in the correlation coefficient of the current signal  
This method is based on correlation coefficients calculated using the traditional correlation formula for two half-cycles of the same polarity current signal. Only three-phase currents must be measured for this method to work. The magnitude of correlation coefficients is compared to a preset threshold to detect fault conditions [14].

The Table 1 shows the comparison of fault detection time of various detection approaches and new method based on SSC signal. From this table it is confirmed that the new method is faster compared to other methods that are existing today.

TABLE I  
COMPARISON OF NEW METHOD WITH EXISTING METHODS

| Methods                                 | Fault detection time (ms) |
|-----------------------------------------|---------------------------|
| Sample-to-sample comparison method      | 7.35                      |
| Moving sum method                       | 6.86                      |
| Discrete wavelet transform based method | 4.71                      |
| Correlation coefficient based method    | 5.12                      |
| New method based on SSC signal          | 4                         |

## V. CONCLUSION

The summation of squared three-phase currents and the moving average approach are used in this research to evaluate a protection strategy for detection of faults in transmission lines. By examining only a single signal (SSC), this technique reduces fault identification time and computational effort. CT and PT take up a lot of room and are also costly. There is no requirement for PT in the fault detection side because we only need CT for current measurement. In normal conditions, the SSC signal has a constant magnitude. However, in the event of any form of disruption, it shows huge changes in magnitude. The amount of variations changes depending on the type of fault. The LLLG fault has the most variation in SSC signal, followed by LLG, LL, and LG fault conditions. The FDC is the ratio of the moving average of SSC signal in the  $n$ th window to the  $(n - 1)$ th window. In abnormal conditions, its value will be more than 1. The FDC value for the LLLG fault is also high because it is the most serious fault in the power system.

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